Christ College Of Engineering
(Autonomous)

Department of Computer Science and Engineering

Software Requirements Specification

(SRS)

EV Charging Slot Booking System

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Contents

1 Introduction **3**

1.1 Purpose..... 3

1.2 Scope 3

1.3 Definitions, Acronyms, Abbreviations 3

1.4 References..... 3

2 Glossary **3**

3 Overall Description 4

3.1 Product Perspective 4

3.2 User Classes 4

3.3 Operating Environment 4

4 System Models 5

4.1 Use Case Diagram 5

4.2 Class Diagram..... 6

4.3 Sequence Diagram 6

5 Use Case Specifications 7

5.1 Use Case UC-01:UserRegistration..... 7

5.2 Use Case UC-02:UserLogin..... 8

5.3 Use Case UC-03:SelectVehicleType..... 8

5.4 Use Case UC-04:EnterLocation..... 8

5.5 Use Case UC-05:ViewChargingStations..... 8

5.6 Use Case UC-06:CheckSlotAvailability..... 9

5.7 Use Case UC-07:BookChargingSlot..... 9

5.8 Use Case UC-08:ManageStations(Admin)..... 9

6 Functional Requirements **9**

7 Non Functional Requirements **10**

8 External Interface Requirements **10**

8.1 User Interface..... 10

8.2 Hardware Interface..... 10

8.3 Software Interface..... 10
8.4 Communication Interface..... 11

9 Requirements TraceabilityMatrix 11

10 Appendix 12

1 Introduction

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of the **EV Charging Slot Booking System**.

This document serves as a reference for developers, project supervisors, and stakeholders..

1.2 Scope

The system allows users to register, log in, select vehicle type, enter location details, view nearby EV charging stations, check slot availability, and book charging slots.

The system ensures that a time slot once booked cannot be used by another user at the same time.

1.3 Definitions, Acronyms, Abbreviations

Term	Description
SRS	Software Requirements Specification
UML	Unified Modeling Language
DBMS	Database Management System
API	Application Programming Interface

1.4 References

- IEEE 830– SRS Standard
- IEEE 29148 – Requirements Engineering
- DBMS and Software Engineering course materials

2 Glossary

Term	Meaning
User	External entity interacting with system
Admin	Person managing stations and slots
Slot	A fixed time interval for charging

3 Overall Description

3.1 Product Perspective

The EV Charging Slot Booking System is a web-based application that connects users with EV charging stations.

The system stores information about:

- Users
- Charging Stations
- Charging Slots
- Bookings
- Vehicle Types

3.2 User Classes

User Type	Description
Admin	Full system control
User	Limited access
Guest	View only access

3.3 Operating Environment

- OS: Windows / Linux
- Database: MySQL 8.0.41
- Backend: Flask
- Frontend: React
- Interface: Web browser

4 System Models

4.1 Use Case Diagram

Actors:

- User
- Admin

User Actions:

- Register
- Login
- Select vehicle type
- Enter location
- View charging stations
- Check slot availability
- Book slot
- Cancel booking

Admin Actions:

- Add station
- Update station
- Manage slots
- View bookings

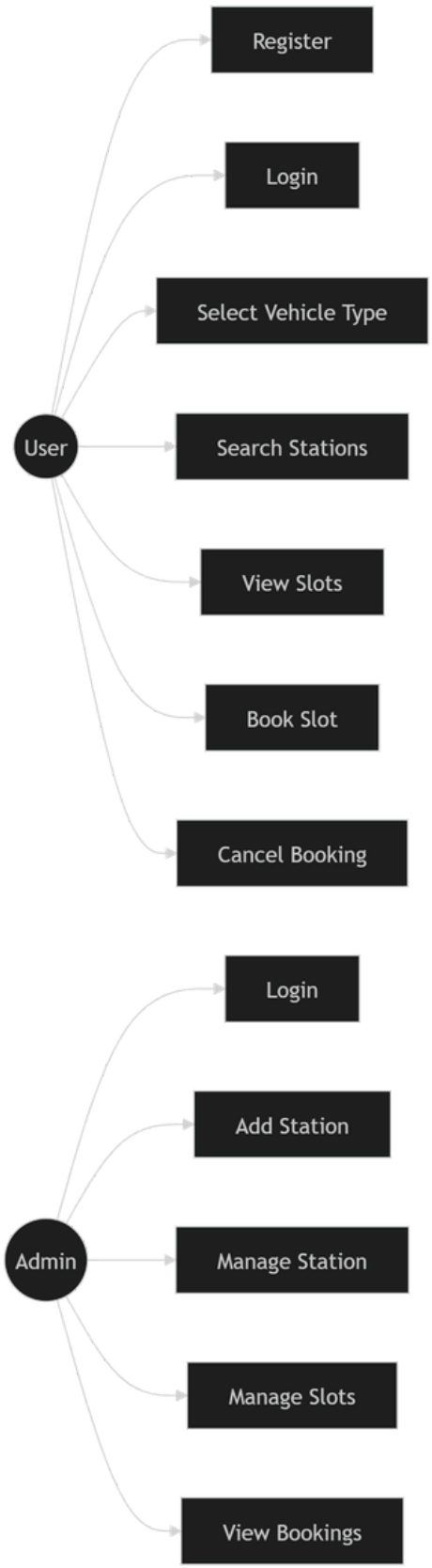


Figure 1: System Use Case Diagram

4.2 Class Diagram

Main classes in the system:

- User
- Vehicle
- Station
- Slot
- Booking

Relationships:

- User makes Booking
- Station contains Slots
- Slot belongs to Station
- Booking is linked to Slot

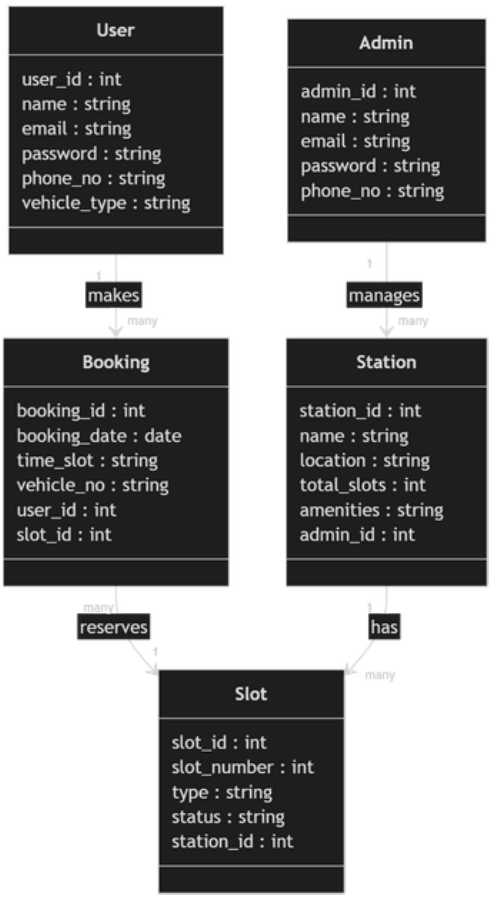


Figure 2: Class Diagram

4.3 Sequence Diagram

Examplesequencefor getting movie recommendation

1. User logs into system
2. User enters location and destination
3. System displays nearby charging station
4. User selects a station
5. System shows available slots

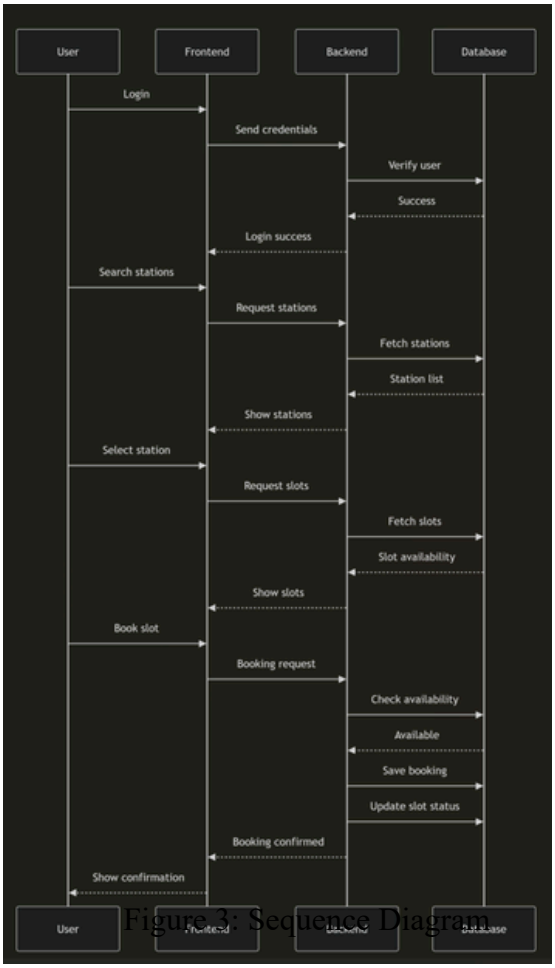


Figure 3: Sequence Diagram

5 Use Case Specifications

5.1 Use Case UC-01: User Login

Use Case ID	UC-01
Use Case Name	User Registration
Primary Actor	User
Description	Allows a new user to create an account in the system.
Precondition	User must not already have an account
Postcondition	Anewuser account is created and stored in the database.

5.2 Use Case UC-02: User Login

Use Case ID	UC-02
Use Case Name	User Login
Primary Actor	User
Description	Allows registered users to login to the system.
Precondition	User must be registered.
Postcondition	User dashboard is displayed.

5.3 Use Case UC-03: Select Vehicle Type

Use Case ID	UC-03
Use Case Name	Vehicle type selection
Primary Actor	User
Description	User selects 2-wheeler or 4-wheeler
Precondition	User must be logged in.
Postcondition	Vehicle type is stored

5.4 Use Case UC-04: Enter Location

Use Case ID	UC-04
Use Case Name	Enter user location
Primary Actor	User
Description	User enters current location and destination.
Precondition	User must be logged in
Postcondition	Location data is processed

5.5 Use Case UC-05: View Charging Stations

Use Case ID	UC-05
Use Case Name	View charging stations
Primary Actor	User
Description	Displays nearby charging stations.
Precondition	Location must be provided
Postcondition	List of stations is displayed

5.6 Use Case UC-06: Check Slot Availability

Use Case ID	UC-06
Use Case Name	Check available slots
Primary Actor	User
Description	Displays available time slots
Precondition	Station must be selected
Postcondition	Slot availability is shown

5.7 Use Case UC-07: Book Charging Slot

Use Case ID	UC-07
Use Case Name	Booking slots
Primary Actor	User
Description	User books a time slot
Precondition	Slot must be available
Postcondition	Slot is reserved and cannot be booked by others.

5.8 Use Case UC-08: Manage Stations (Admin)

Use Case ID	UC-08
Use Case Name	Manage Stations (Admin)
Primary Actor	Admin
Description	Admin adds or updates charging stations
Precondition	Admin must be logged in
Postcondition	Station data is updated

6 Functional Requirements

FR-01: System shall allow user registration

FR-02: System shall authenticate users

FR-03: System shall allow vehicle type selection

FR-04: System shall allow location input

FR-05: System shall display nearby charging stations

FR-06: System shall show slot availability

FR-07: System shall allow slot booking

FR-08: System shall prevent double booking

FR-09: System shall allow booking cancellation

FR-10: Admin shall manage stations and slots

7 Non Functional Requirements

- **NFR-01: Response time should be less than 3 seconds NFR-**
- **02: System must ensure secure authentication NFR-03:**
- **System availability should be at least 99% NFR-04: Interface**
- **must be user-friendly NFR-05: System must handle multiple**
- **users simultaneously**

8 External Interface Requirements

8.1 User Interface

1. Web-based UIfor searching and booking slots
2. Map or list view for stations

8.2 Hardware Interface

The system runs on standard computing devices such as:

- Desktop computers
- Laptops
- Mobile devices

8.3 Software Interface

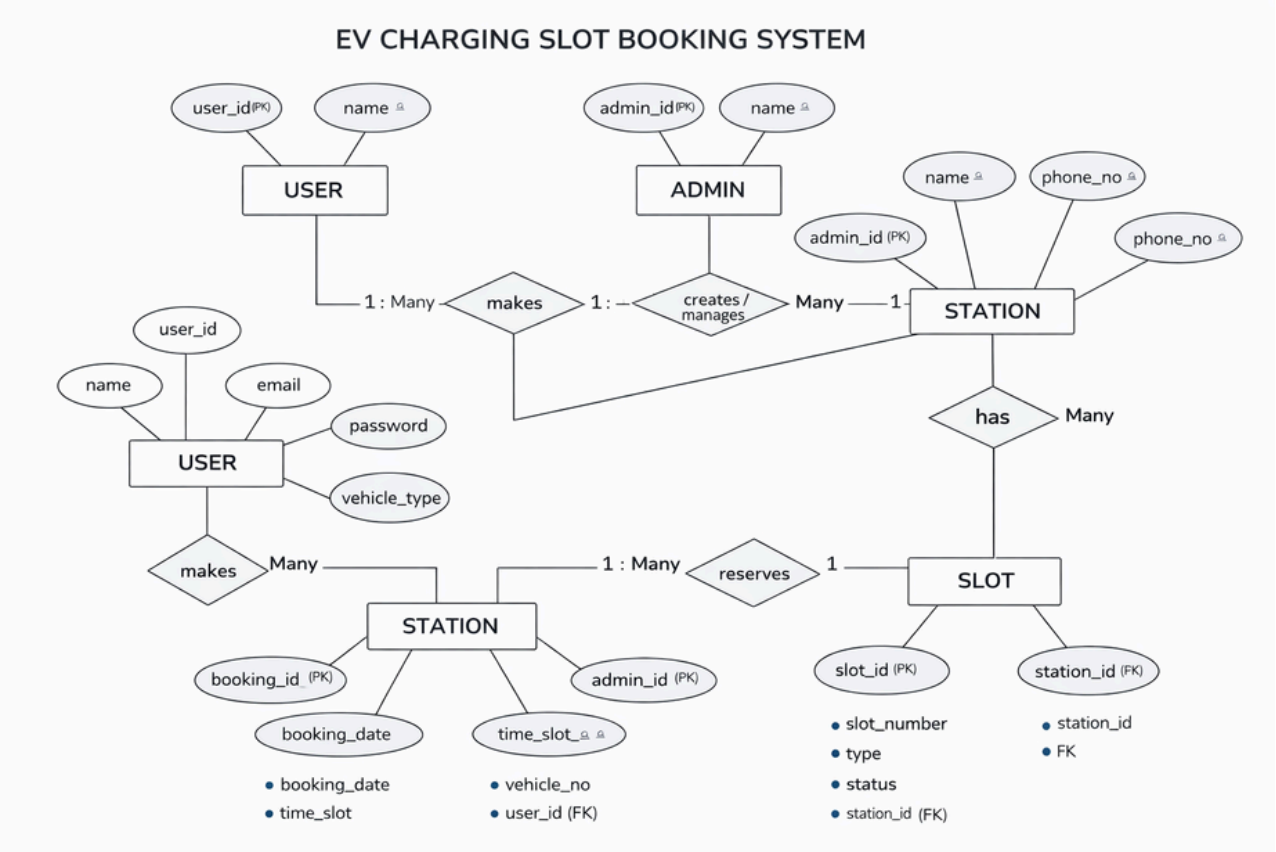
- MySQL DatabaseServer
- Web Browser
- Backend server

8.4 Communication Interface

- HTTP protocol
- RESTful API

9 Requirements Traceability Matrix

Requirement ID	Design Module	Test Case
FR-01	Registration Module	TC01
FR-02	Login Module	TC02
FR-03	Vehicle Selection Module	TC03
FR-04	Location Input Module	TC04
FR-05	Station Display Module	TC05
FR-06	Slot Availability Module	TC06
FR-07	Booking Module	TC07

10 Appendix

- ER Diagram